MISCELLANEOUS — FULL MIND MAP

Chemistry · Chapter 11 · Topic-wise · Fully Explained · Hinglish · SSC/PCS/UPSC Ready

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1. Nobel Prizes, Special Glass, HFC–CFC & Ozone

- 2021 ka Chemistry Nobel Benjamin List aur David MacMillan ko asymmetric organocatalysis ke development ke liye mila; 2019 ka Nobel John Goodenough, Stanley Whittingham aur Akira Yoshino ko lithium-ion battery ke development ke liye diya gaya tha. [Q1–Q2]
- Crookes glass UV rays ki transmission ko reduce karta hai aur sunglasses me use hota hai; is glass me cerium oxide jaise additives silica ke saath use kiye ja sakte hain. [Q3]
- HFCs ka use refrigeration/AC, aerosol propellants, insulating foams aur fire-protection applications me hota hai; source question me aerosols + foam agents + fire retardants = teen categories correct hain, lubricants nahi. [Q4]
- Stratospheric ozone depletion ka major cause CFCs se release hone wale chlorine/bromine radicals hain. Ozone water me microorganisms ko apni strong oxidizing power se kill karta hai; uska non-radioactive hona reason nahi hai. [Q5–Q6]

2. Fire Safety: Alarm, Extinguishers & Fireproofing

- Photocell/photoelectric sensor fire-alarm systems me smoke se light path change detect karne ke liye use ho sakta hai. [Q7]
- CO_2 fire extinguisher oxygen supply ko displace karke combustion ko suppress karta hai. Traditional soda-acid extinguisher me sodium bicarbonate solution aur dilute H_2SO_4 react karke CO_2 generate karte hain. [Q8–Q10]
- Source ke according Aluminium sulphate textile treatment me fireproof clothing banane ke liye use kiya jata hai. [Q11]

3. Repellents, Rodenticides, Pesticides & Herbicides

- Mosquito control me Allethrin aur Prallethrin synthetic pyrethroids hain; Pyrethrum/Pyrethrins Chrysanthemum-based natural insecticidal compounds hain, jabki Lemongrass/Citronella oil natural repellent ke roop me use hota hai. [Q12–Q16]
- Rodent-control questions me source Potassium cyanide aur Zinc phosphide ko rat poison/rodenticide se link karta hai; “rodenticide” specifically rats/mice jaise rodents ko control karne wale chemicals ke liye term hai. [Q17–Q19]
- Aluminium phosphide stored grains ke fumigation me phosphine release karta hai aur rodent-control use bhi mil sakta hai, isliye source ne is question ko multiple-correct maana hai. [Q20]
- Herbicides/weedicides me Sodium chlorate non-selective herbicide hai; 2,4-D selective systemic herbicide/synthetic auxin hai aur paddy me broadleaf weeds ke control se linked hai; Butachlor rice cultivation ka common herbicide hai. [Q21,Q23–Q24,Q26]
- Carbofuran, methyl parathion, phorate aur triazophos agriculture pesticides hain jinke toxicity/environmental risk ko lekar regulation concern raha hai. DDT persistent, non-biodegradable pollutant hai jo food chain me accumulate karta hai. [Q22,Q25]

⚠ EXAM TRAP — Q20: Aluminium Phosphide ka ek hi use nahi

- ▸ Stored-grain fumigation aur rodent-control dono context milte hain, isliye source ne Ans. * (multiple correct) diya hai.
- ▸ Is compound ki toxicity bahut high hai; exam me sirf functional classification yaad rakho, practical use/handling details nahi.

4. Radioactive Dating, Carbon Credit, Isotopes & Geoengineering

- Carbon-14 dating once-living materials—wood, bone, shell, fossils—ki age estimate karne ke liye use hoti hai. Rocks/geological age ke liye radioactive dating me U-Pb ya K-Ar jaise methods use hote hain; Earth ki age ke context me Uranium process accepted hai. [Q27–Q31]
- 1 carbon credit generally 1 metric ton = 1000 kg CO_2-equivalent emission right/reduction ko represent karta hai. BRIT (Board of Radiation and Isotope Technology) DAE ke under isotope/radiation technology services se linked hai. [Q32–Q33]
- Zinc oxide ko “philosopher’s wool” kaha jata hai; urinals ke paas pungent smell urea ke bacterial breakdown se banne wale ammonia ki wajah se hoti hai. [Q34–Q35]
- Antihelium-4 jaise antimatter nuclei universe evolution aur antimatter stars/galaxies ki possibility ko probe karne me useful hain, mineral prospecting ko cheap banana iska practical implication nahi hai. [Q36]
- Cirrus-cloud thinning aur stratosphere me sulphate aerosol injection proposed geoengineering techniques hain jinka aim planetary cooling/global warming ko reduce karna hai. [Q37]

5. Artificial Rain, Common Chemicals & Match-the-Following

- Cloud seeding/artificial rain ke liye Silver iodide sabse common exam answer hai; dry ice bhi use ho sakta hai. Rainfall potential atmospheric humidity/moisture par strongly depend karta hai. [Q38–Q41]

- Key matching yaad rakho: Blue vitriol ($CuSO_4$) → fungicide; Eosin → red ink; Silver iodide → artificial rain; Zinc phosphide → rodenticide. Zinc phosphide–rat poison correct pair hai. [Q42–Q43]
- Kevlar → synthetic fibre, Taxol → anticancer drug, Zinc phosphide → rodenticide, Nitrocellulose → explosive. Silver bromide photography me use hota hai, artificial photosynthesis me nahi. [Q44–Q45]
- Silicon carbide → artificial diamond/abrasive context; Carbon fibre → aircraft; CO_2 → photosynthesis; dichlorodifluoromethane/Freon → refrigerator. [Q46]
- Iron → haemoglobin; Lead → storage battery; Silver compounds → photography; Copper → lightning conductor. Morphine → analgesic; Sodium → kerosene storage; Boric acid → antiseptic; German silver → alloy. [Q47–Q48]
- Skin cancer ↔ UV exposure, noise pollution ↔ decibel, global warming ↔ CO_2, ozone hole ↔ CFCs—ye standard environmental matching hai. [Q49]

6. Metals, Pigments, Solvents, Phosphorus, CNT & Desalination

- Titanium ko “metal of future” kaha jata hai kyunki ye light, strong aur corrosion-resistant hai. Electric bulb filament Silver ka nahi, Tungsten ka hota hai because tungsten ka melting point bahut high hai. [Q50–Q51]
- Vermillion chemically Mercuric sulphide / Mercury(II) sulphide (HgS) hai. Traditional dry-cleaning me benzene, petrol/kerosene, alcohol jaise solvents historically use hue; modern systems me perchloroethylene common raha hai. [Q52–Q54]
- Red ink ke liye Eosin dye use hota hai. Source Q56 Sodium chromate ko red colour se associate karta hai (standard chemistry caveat niche trap me hai). Fireworks me Strontium salts bright crimson/red colour dete hain. [Q55–Q57]
- Acetone → nail-polish remover; Carbon tetrachloride → historical fire-extinguisher use; H_2O_2 → wound dressing/antiseptic context; liquid ammonia → refrigerant. Accumulator/car battery me HCl nahi, H_2SO_4 electrolyte hota hai. [Q58–Q59]
- Safety matches me phosphorus especially Red phosphorus striking surface se linked hai; White phosphorus air aur darkness me chemiluminescence ki wajah se glow karta hai. [Q60–Q62]
- Carbon nanotubes ki modern discovery/characterisation Sumio Iijima (1991) se linked hai. India me desalination plants Lakshadweep aur Chennai dono jagah milte hain; first 1-lakh-LPD LTTD plant Kavaratti me commission hua. [Q63–Q65]
- Alkaline soil reclamation ke liye Gypsum/Calcium sulphate use hota hai; “edaphic” term soil-related factors ke liye hai. Very-low-temperature technology, jisme liquid O_2/H_2 handle hote hain, Cryogenics kehlati hai. [Q66–Q68]

⚠ EXAM TRAP — Q56 Sodium Chromate colour

- Source key Sodium Chromate ko “Red” se associate karta hai, lekin sodium chromate crystals/solutions generally yellow hote hain.
- Isliye is fact ko source-key specific samjho; standard chemistry me chromate = yellow aur dichromate = orange yaad rakhna safer hai.

7. Chemical Warfare, Smoke Screen, Agent Orange & Nuclear Basics

- Mustard gas (sulphur mustard) WWI me chemical-warfare agent ke roop me use hua. Naam “gas” hai, lekin room temperature par ye oily/viscous liquid ho sakta hai aur aerosol/mist ke form me disperse kiya gaya. [Q69–Q72]
- Military smoke screens dense particulate clouds create karte hain; source me titanium oxide particles/$TiCl_4$ moisture reaction ka context diya hai, aur zinc chloride bhi smokescreen chemistry se linked hai. [Q73]
- Agent Orange Vietnam War me US military ka defoliant/herbicide tha; severe health effects ka major concern dioxin (TCDD) contaminant se linked tha. [Q74–Q75]
- Nucleus ka split hona nuclear fission hai; multiple light nuclei ka combine hokar heavier nucleus banana fusion hai. Isliye “breaking apart = fusion” statement incorrect hai. [Q76]

8. Cells, Batteries, Microwave & Everyday Chemical Applications

- Common zinc-carbon dry cell me $NH_4Cl + ZnCl_2$ paste electrolyte hota hai; ye primary/non-rechargeable cell hai because reactions readily reversible nahi hoti. Stored chemical energy discharge par electrical energy me convert hoti hai. [Q77–Q81]
- Lead-acid car battery ka electrolyte dilute H_2SO_4 hai. Rechargeable Ni-Cd batteries me Nickel aur Cadmium electrode materials hote hain, KOH electrolyte context milta hai. [Q82–Q86]
- Microwave oven me high-frequency microwaves generate karne ke liye Magnetron tube use hoti hai. [Q87]
- Freon → refrigerant; Tetraethyl lead → anti-knocking agent; BHC → insecticide; Carbon tetrachloride → historical fire extinguisher. Aspartame → artificial sweetener; Neoprene → synthetic rubber; Benadryl → antihistamine. [Q88–Q89]
- Source matching ke according lipstick–lead, soft drink–brominated vegetable oil aur Chinese fast food–MSG tino pairs accepted hain. Potassium bromide → photography; potassium nitrate → gunpowder; potassium sulphate → fertilizer; monopotassium tartrate/cream of tartar → bakery. [Q90–Q91]
- Ozone depletion–skin cancer correct environmental pair hai. Household phenyl germicidal action phenol/cresol derivatives se linked hai. [Q92–Q93]
- Source matching: Rutherford → atom bomb, Alfred Nobel → dynamite, Cartwright → powerloom, Graham Bell → telephone. Important dates: Technology Day 11 May, Photography Day 19 Aug, National Sports Day 29 Aug, Nagasaki Day 9 Aug. 2011 ko UN ne International Year of Chemistry designate kiya tha. [Q94–Q96]

⚠ EXAM TRAP — Q88 & Q94 source simplifications

- CCl_4 ka fire-extinguisher use historical hai; toxicity ke karan modern fire safety me ye standard choice nahi hai.
- “Rutherford → Atom Bomb” source matching oversimplified hai. Rutherford nuclear physics pioneer the; atomic-bomb project ka “father” label commonly J. Robert Oppenheimer se associated hai.

9. Climate Agreements, Greenhouse Gases, Air Pollution & Hydrogen

- UNFCCC ka focus greenhouse-gas emissions ko stabilize/mitigate karna hai, isliye CO_2 reduction bhi naturally scope me aata hai. Alma-Ata → primary healthcare aur Talanoa Dialogue → global climate action correct pairs hain; Hague Convention/Under2 Coalition ke source pairs incorrect hain. [Q97–Q98]
- Brazil ne long-standing ethanol–gasoline blending mandate adopt kiya. Air-pollutant question me HCN aur H_2S dono inorganic gaseous pollutants ho sakte hain; H_2S ka major natural/environmental source decaying vegetation/animal matter aur anaerobic decomposition hai. [Q99–Q101]
- CO_2, CH_4 aur N_2O greenhouse gases hain. Source commission estimate me relative contribution order CO_2 > CH_4 > N_2O > CFCs diya gaya hai. Atmospheric water vapour altitude ke saath decrease karta hai aur poles par maximum nahi, warm humid tropics me zyada hota hai. [Q102–Q104]
- Nuclear/thermal power plants ke heated cooling-water discharge se thermal pollution ho sakta hai. "Green Muffler" dense vegetation barrier se noise pollution reduce karne ki technique hai. [Q105–Q106]
- Urban magnetic pollution nanoparticles vehicle brakes, engines/tailpipe combustion aur power plants se aa sakte hain. Petrol/diesel ki incomplete combustion Carbon monoxide aur soot produce karti hai. [Q107–Q108]
- Metropolitan air me CO, NO_x, SO_2, ozone, lead aur particulate matter jaise multiple pollutants present hote hain; isi wajah se source ne Q109 ko disputed maana. PM ek air pollutant hai. [Q109–Q110]
- Source-era RSPM question me <5 μm accepted answer hai, jabki modern monitoring PM_{10}/$PM_{2.5}$ categories extensively use karti hai. NAMP regular monitoring me SO_2, NO_2, PM_{10} aur $PM_{2.5}$ major parameters hain. [Q111–Q112]
- Steam methane reforming se, jab CO_2 capture na ho, "Grey Hydrogen" banta hai. Green Hydrogen fertilizer plants, oil refineries aur steel plants tino ko decarbonize karne me role play kar sakta hai; direct combustion, natural-gas blending aur fuel-cell vehicles tino use-cases possible hain. [Q113–Q115]

⚠ EXAM TRAP — Q100, Q103 & Q109

- Q100 me HCN aur H_2S dono technically inorganic gaseous pollutants hain, isliye source multiple-correct maanta hai.
- Q103 ka greenhouse-gas contribution order commission-specific estimate hai; "global warming potential" aur "total contribution" ko mix mat karo.
- Q109 me metropolitan pollution ke multiple options valid the, isliye source answer * (disputed) hai.

10. Smog, Water Pollution, Waste Treatment & Mercury

- Classical/industrial smog me nitrogen aur sulphur oxides, smoke/particulates important hote hain. Photochemical smog sunlight + NO_x + hydrocarbons se banta hai aur Ozone iska key secondary pollutant hai. [Q116–Q117]
- Eutrophication water-body ka nutrient enrichment hai, especially nitrogenous nutrients/nitrates aur phosphates/orthophosphates se. Algal bloom ke decomposition se dissolved oxygen deplete hota hai. [Q118–Q119]
- BOD aquatic pollution ka standard indicator hai: organic matter decompose karne me microorganisms jitna oxygen consume karte hain, BOD utna high; high BOD generally higher organic pollution ko show karta hai. [Q120]
- Industrial/sewage effluents water ke pH, temperature aur chemical composition jaise physico-chemical characteristics change karte hain. Membrane Bioreactor (MBR) wastewater-treatment technology hai jo biological treatment + membrane separation combine karta hai. [Q121–Q122]
- Pyrolysis aur plasma gasification waste-to-energy thermal technologies hain jo waste ko high temperature par syngas/bio-oil jaise useful products me convert kar sakte hain. [Q123]
- Mercury pollution ke sources me historical mercurial pesticides, dental amalgam, fluorescent lamps aur coal-based thermal power plants sab include ho sakte hain. [Q124]

✓ Source-question mapping preserved: Q1–Q124